

Material Science

Composite Materials: Bridging Theory, Experiment and Innovation

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1. Introduction

This assignment explores composite materials through the application of composites within Formula One (F1). The goal of this assignment is to devise a piece of composite material for Crimson Arrow Racing to improve the power to weight ratio of the race car. This will be compared to real pieces of composite materials, that have got supported physical evidence of. Theoretical and empirical analysis of the real composites can be compared to the proposed new composite to validate the material and show its potential usefulness in this application.

Composite materials are materials made from two or more constituent phases, typically a matrix and a reinforcement (such as fibres), when non atomically bonded produce a material superior to that of the components individually. The matrix binds the fibres together, whilst the fibres provide the strength and stiffness. This will give a material that is tailored to a specific goal. [1]

Formula One cars today are majority carbon fibre. Almost 80% of the cars are carbon fibre. Carbon fibre reinforced polymer (CFRP) is used in the monocoque chassis. The monocoque chassis (seen in figure 1) is the main protection for the driver if in an impact occurs. CFRP is about five times as strong as steel but much lighter. Carbon fibre isn't only used in the main structure, but it is used for brake discs and pads. This is also due to carbons brilliant thermal capabilities as well as being very lightweight. These brakes can be subject to temperatures over 1000 °C during a race in heavy braking zones. [2] Carbon fibres are used throughout F1 to reduce weight but also maintain structural integrity. The power to weight ratio is critical in Formula One as it can massively impact acceleration, handling, and fuel efficiency.

Recent research in composites has been aimed at improving matrix bonding with the reinforcement, improving recyclability and development of reinforcement materials like 3D graphene and bio based fibres. In formula one this can be very useful as development is always aiming at optimising power to weight ratio without sacrificing on the strength of a material [3]

The report will cover a summary of composite structure and properties, new composite material for the application mentioned above. Theoretical calculations and analysis and experimental data analysis with a comparison between proposed single ply and ply that physical evidence has been collected on.

Figure 1: Carbon fibre within motorsport applications

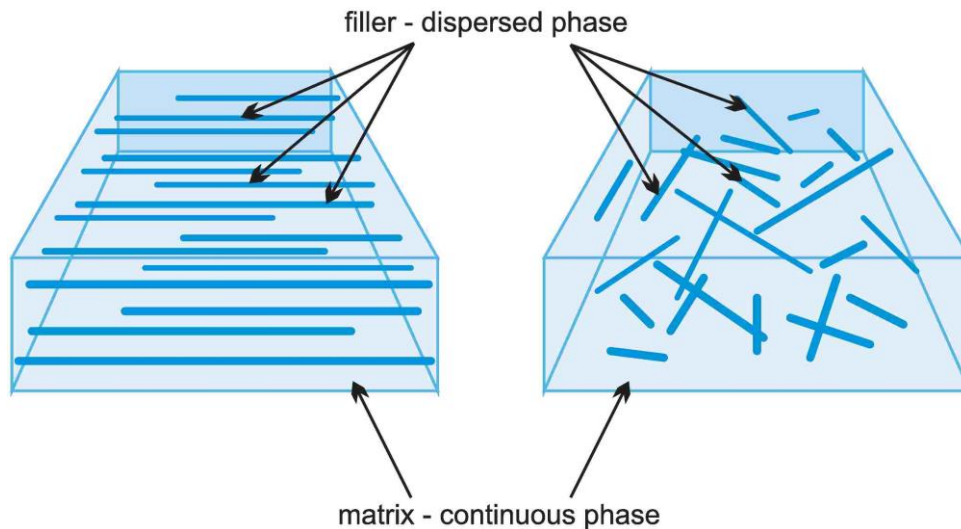


2. Composite structures and properties

Matrix is the continuous phase that surrounds and supports the fibres, as seen in figure 2. It binds them together and transfer load between them. Key roles include load transfer, between fibres and even distribution of stress. Protection, shields fibres from environmental damage. Structure, it holds the fibres in place and maintains the overall shape of the composite. [4] Common Matrix materials include polymers (epoxy and polyester), metals (aluminium and titanium), and ceramics (Silicon dioxide and alumina.) [5]

Fibres are the dispersed phase (as seen in figure 2) and are primarily responsible for the strength and stiffness of the composite. Key roles include strength and stiffness, provide high tensile strength and rigidity. Load bearing as fibres carry the majority of the load on a composite. Directional Properties as fibres can be oriented to tailor for mechanical properties e.g. unidirectional or woven. [6] Common fibre materials are glass fibres, carbon fibres, aramid fibres and natural fibres. [7]

Figure 2: Illustration showing the phases of a composite

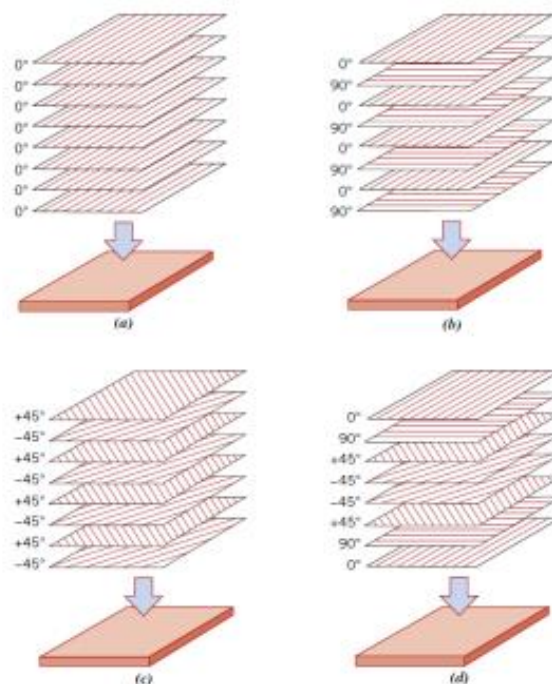


The interaction between the matrix and the fibres contribute massively to mechanical performance. Firstly, load transfer is determined by the interaction between matrix and fibre. The fibres carry most of the load, but they can only transfer it if the matrix distributes the stress evenly to the fibres. Strong interfacial bonding ensures that when a load is applied to a composite the load moves from the matrix to the fibres without slippage. Poor bonding can lead to debonding and therefore a reduce in strength and stiffness. If fibres are discontinuous or oriented differently the matrix can distribute the stress and prevent stress concentrations which could lead to failure.

Another way in which they contribute to mechanical performance is crack arrest and damage tolerance. When cracks form in the matrix, fibres can act and stop cracking propagation. On the other hand, the matrix prevents fibre buckling and keeps the fibres aligned in the orientation. Also, good adhesion means cracks are deflected rather than travelling straight through the composite therefore improving fracture toughness. As mentioned previously the interaction could lead to have environment and fatigue resistance. The matrix protects fibres from moisture, chemicals and UV wear and a strong fibre matrix bonding reduces micro cracking under load and reduces fatigue. Engineers and designers can also tailor a composite in a specific way depending on the application. Designers could give the composite an anisotropic structure these means high stiffness in one direction but flexibility in another direction. [8] [9] [10]

Laminates can be formed in multiple difference ways, seen in figure 3. Unidirectional laminates, this is where all the fibres align in the same direction. This provides maximum strength and stiffness along that axis however will have poor properties on the other axis, this makes them highly anisotropic. Another layout is cross ply laminates. Plys alternate between 0° and 90° , this improves strength and stiffness in those two directions and leads to better dimensional stability. Angle ply laminates with alternating $\pm 45^\circ$ orientations. This is ideal for improving shear stiffness and torsional strength. Multi directional laminate uses various orientations (e.g. 0° , $\pm 45^\circ$, and 90°) to provide a balanced stiffness and strength in all directions. A design could so incorporate woven laminates. Woven uses interlaced fibres, which improves impact resistance and mouldability, although tensile properties are decreased through this process. Engineers and designers will weigh up each method depending on the application requirements. [11]

Figure 3: Different ply structures seen in composites
 (a) Unidirectional ply, (b) cross ply, (c) angle ply, (d) multidirectional ply



3. Material selection for a single ply

Application – Formula One Front Wing

Matrix – Epoxy Resin

Epoxy resin has superior bonding adhesion to carbon fibres; this allows for more efficient load transfer. It also has excellent strength properties better than polyester or vinyl ester. It has low shrinkage and high dimensional accuracy which is key for aerodynamic parts. Can withstand heat from the track, tyre and brakes without degrading and suits prepreg and autoclaving manufacturing processes which are common throughout F1. [12]

Fibre – M40J Carbon Fibre

M40J has a very high modulus (~377 GPa), which means it has excellent tensile strength (~4,400MPa). It also has low density (~1.8 g/cm³) [13], this keeps the wing lightweight, therefore improves the cars performance and reducing inertia. It maintains its shape under high temperatures and mechanical load, ensuring consistent aerodynamic performance. This can vary slightly depending on number of fibres. [14]

Alternative Options (Matrix) –

Alternative matrix options could be polyester resin and vinyl ester resin. Polyester resin is less expensive and more sustainable however, it has inferior mechanical properties and durability compared to epoxy resin, making it unsuitable for the application. Another option is Vinyl Ester resin, this resin has lower cost, but slightly more than polyester and it has similar sustainability to polyester. Although Vinyl Ester offer slightly less favourable mechanical properties for our application this resin is worth considering for future applications.[15]

Alternative Options (Fibre) –

One possible alternative fibre could be T700. T700 has a higher tensile strength than M40J, also T700 has significant cost savings compared to M40J. On the other hand, T700 offers a much lower modulus than M40J, which mean T700 will have more deflection under a load which is a crucial consideration for f1 front wings also M40J is 0.03 g/cm³ less dense, making this fibre less suitable for this application. [16] [17] A future consideration for an alternative is 3D graphene. 3D graphene builds on 2D graphene properties like high strength and high thermal conductivity and adds super low density and improved all round mechanical properties. This is a new solution that can be found in forms such as foams and aerogels. It is reported that density such as 0.01 to 0.1 g/cm³ are found in these forms. Graphene in its current form has a tensile Strength ~ 130 GPa, which is currently less than M40J however with the advancements in 3D graphene this could easily increase. [18] [19] Both of these options are not as suitable for F1 front wings as M40J but offer good current and future alternatives.

Mechanical Performance –

Extremely high stiffness, which is good for aerodynamic stability and minimum deflection, this is a key part of formula one front wing design as in Article 3.15.4 of the F1 regulations states, when the load is applied symmetrically to both sides of the car, the vertical deflection must be no more than 10mm, and when applied to only one side of the car, the vertical deflection must be no more than 15mm.[20] This is therefore a key characteristic and the material suits this well. The resin gives high adhesion, chemical and thermal resistance. The drawback to this composite is the brittleness. [21]

Cost-effectiveness –

M40J costs \$59–\$67 per linear yard which is \$130–\$150 per kg of composite material. [16] Epoxy resin is relatively affordable (\$20-\$40 for 500ml) [22], the more expensive part is the manufacturing of these parts. In F1 prepreg and autoclave are used to manufacture these parts. Prepreg is where composite fibres where the matrix material is already present. The fibres are often in a form of a weave, and the matrix is used to bond them together during manufacturing. After prepreg matrices are left partially cured to allow for easier handling. An autoclave is then required to cure this composite. [23] An autoclave is a high pressure, high temperature vessel that cures composite materials. [24]

Sustainability and recyclability –

Carbon fibres used for this function don't have great sustainability and recyclability. This due to the high costs and high energy to recycle them. Lots of research is ongoing to improve the sustainability of carbon fibres. A recent innovative alternative for recycling carbon fibres is called solvolysis. This process involves using a solvent to break down the matrix and the fibres but requires lots of energy (~260MJ/kg). [25] There is room for improvement however research is being done to do so. [26]

Material Property Data –

Table 1: M40J Carbon Fibre Properties: [12] [27]

Mechanical Property	Value
Tensile Strength	4,400 MPa
Tensile Modulus	377 GPa
Density	1.75 g/cm ³
Elongation	1.2%

Table 2: Epoxy Resin (thermoset epoxy): [28]

Mechanical Property	Value
Tensile Strength	100 MPa
Tensile Modulus	3300 GPa
Typical glass transition temperature (T _g)	180 °C

Table 3: Composite Properties (M40J/EP): [29]

Mechanical Property	Value
Tensile Strength	2,500 MPa
Tensile Modulus	226 GPa
Tensile Strain	1.11%
Compressive Strength	1,270 MPa
Flexural Strength	1,560 MPa
Flexural Modulus	190 GPa
ILSS (Interlaminar Shear Strength)	83.1 MPa
In-Plane Shear Strength	123 MPa
90° Tensile Strength	63 MPa
Fibre volume	60%

$$\text{Volume Fraction} = \frac{\text{Volume of Constituent}}{\text{Total Volume of composite}}$$

$V_f = \text{Volume of Fibre}$

$V_m = \text{Volume of Matrix}$

where $V_f + V_m = 1$

So, in the composite I have chosen the values are $V_f = 60$ and $V_m = 40$

Specialised Property –

Formula One front wings are designed to be anisotropic; this means they can be stiff in one direction and flexible in another. The FIA deformation tests are done in the vertical direction however in practice the wings are subject to longitudinal forces. This anisotropic idea can be used to pass the FIA test whilst in practice the wings can deform in another direction. [10] This is called Aero-Elastic Behaviour. This can be a controlled amount of flex in the wing. It can be used for decreasing drag in the straights whilst being able to maintain downforce in the corners. This can play an important part on decreasing lap time whilst keeping the aero package on the car unchanged.

4. Theoretical $[E_c]$ calculation – $[0]_{16}$ laminate

Using the data from table 3 the longitudinal modulus can now be found, this is more commonly seen as Voigt Modulus, and the rule of mixtures can be used to find this value. The rule of mixtures is a method used to estimate the properties of a composite based on its individual modulus and its volume ratio of the composite.[30] For this calculation the assumptions are that we have 16 ply's all perfectly at 0° , this means the load is spread evenly throughout the composite. Another assumption is that perfect bonding has occurred, so no slippage or voids are present. A final assumption is that the properties are constant in each section of fibre and matrix across the composite.

$$E_c = V_f E_f + V_m E_m$$

Where:

E_c = Voigt Modulus (Longitudinal modulus of the composite)

V_f = Volume of the fibre

E_f = Fibre Modulus

V_m = Volume fraction of Matrix

E_m = Matrix Modulus

Inserting values from tables 1 and 2:

$$E_f = 377 \text{ GPa}$$

$$V_f = 0.60$$

$$E_m = 3.3 \text{ GPa}$$

$$V_m = 0.40$$

$$E_c = 0.60 \times 377 + 0.4 \times 3.3 = \mathbf{227.52 \text{ GPa}}$$

A calculated Voigt Modulus of **227.52GPa** has been calculated for M40J Epoxy resin composite. The modulus is due to its unidirectional composite structure. As it is 0° , this means that the modulus is parallel to the direction of the load. Due to this high Voigt Modulus, the front wing will be very stiff which means it will have very minimal flex under a high load. These means it will have good aerodynamic consistency. The high Modulus will also mean it pass strict f1 regulations. Overall, this means using this material will lead to a high power to weight ratio whilst maintaining stiffness.

5. Theoretical data analysis

During a lab session three composites were observed during a three point bend test, the aim is to find the modulus of elasticity in the three composites. In this section, the rule of mixtures will again be used alongside the 10% rule to determine the Voigt Modulus. The 10% rule is commonly used for a quick estimate of stiffness and strength. The 10% rule states that off axis ply's ($\pm 45^\circ/90^\circ$) contributes $\sim 10\%$ of the axial strength and stiffness.[31] The same assumptions will be made in the calculation as mentioned in section 4. The properties of the composite used are seen below in table 4, and the differences between the 3 composite ply sequences are as seen in table 5.

Table 4: Sample material properties (all samples have the same properties)

Mechanical Property	Value
E_{fibres}	206GPa
E_{matrix}	3.5GPa
V_{fibres}	0.60
V_{matrix}	0.40

Table 5: Sample composite details

Sample No.	Stacking sequence	Ply subscript
1	16 ply's at 0°	$[0]_{16}$
2	16 ply's at 90°	$[90]_{16}$
3	8 ply's at $\pm 45^\circ$	$[+45/-45]_{4s}$

5.1 Theoretical Calculations for Test 1

Using the properties detailed in table 4, the Voigt Modulus can be calculated:

$$E_c = V_f E_f + V_m E_m$$

$$E_c = 0.60 \times 206 + 0.4 \times 3.5 = 125 \text{ GPa}$$

The Voigt Modulus of the $[0]_{16}$ has been calculated at 125GPa. In this composite all the fibres are aligned parallel to the direction of the load being applied, this is a unidirectional laminate. Since carbon fibres have a high modulus in this direction, this composite can withstand a much higher load. The matrix in this direction will transfer the load between each fibre giving the composite a high Voigt Modulus in this direction.

5.2 Theoretical Calculations for Test 2

$$E_c = u_{(0^\circ)} * E_{0^\circ} + \frac{u_{(90^\circ)}}{10} * E_{0^\circ} + \frac{u_{(45^\circ)}}{10} * E_{0^\circ} + \frac{u_{(-45^\circ)}}{10} * E_{0^\circ}$$

$$E_c = 0 * 125 + \frac{1}{10} * 125 + \frac{0}{10} * 125 + \frac{0}{10} * 125 = \mathbf{12.5GPa}$$

The Voigt Modulus of the $[0]_{90}$ has been calculated at 12.5GPa. In this composite all the fibres are oriented perpendicular to the direction of the load. Therefore, the fibres are contributing very little to the modulus in this off axis direction. This means that the matrix is carrying most of the load. Since the matrix has a much lower stiffness and modulus the overall composite Voigt Modulus drops dramatically.

5.3 Theoretical Calculations for Test 3

$$E_c = u_{(0^\circ)} * E_{0^\circ} + \frac{u_{(90^\circ)}}{10} * E_{0^\circ} + \frac{u_{(45^\circ)}}{10} * E_{0^\circ} + \frac{u_{(-45^\circ)}}{10} * E_{0^\circ}$$

$$E_c = 0 * 125 + \frac{0}{10} * 125 + \frac{0.5}{10} * 125 + \frac{0.5}{10} * 125 = \mathbf{12.5GPa}$$

The Voigt Modulus of the $[+45/-45]_{4s}$ has been calculated at 12.5GPa. In this composite all the fibres are oriented partially in the loading direction. The load should be share between the fibre and matrix however due to a limitation in the 10% rule this is not observed in the calculation.

Table 6: Summary of theoretical results

Sample No.	Stacking sequence	Theoretical Modulus (GPa)
1	$[0]_{16}$	125
2	$[90]_{16}$	12.5
3	$[+45/-45]_{4s}$	12.5

6. Experimental Data Analysis

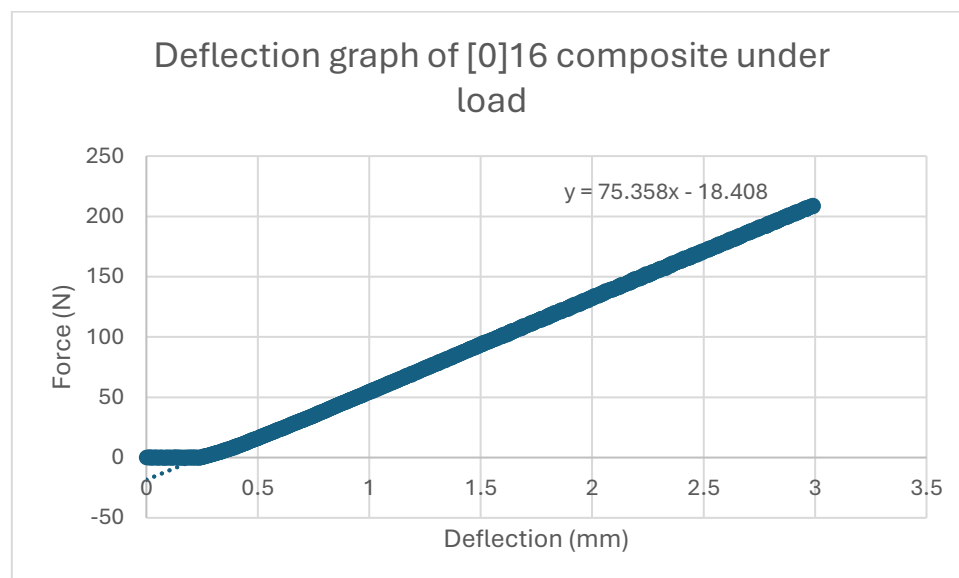
From the laboratory session a 3 point bend test was completed on the composites specified in section 5, table 4 and 5. As a load was on the composites the deflection was measured. This has allowed for a Force, deflection graph to be plotted. From this using:

$$E = slope \cdot \frac{L^3}{4bd^3}$$

The young's modulus can be found for each composite, and therefore we can compare the results to the theoretical data analysis in section 5.

6.1 Experimental Calculations for Test 1

Figure 4: Deflection graph for [0]₁₆



$$E = slope \cdot \frac{L^3}{4bd^3}$$

$$Slope = 75,358 N/m$$

$$L = 0.122m$$

$$b = 0.02592m$$

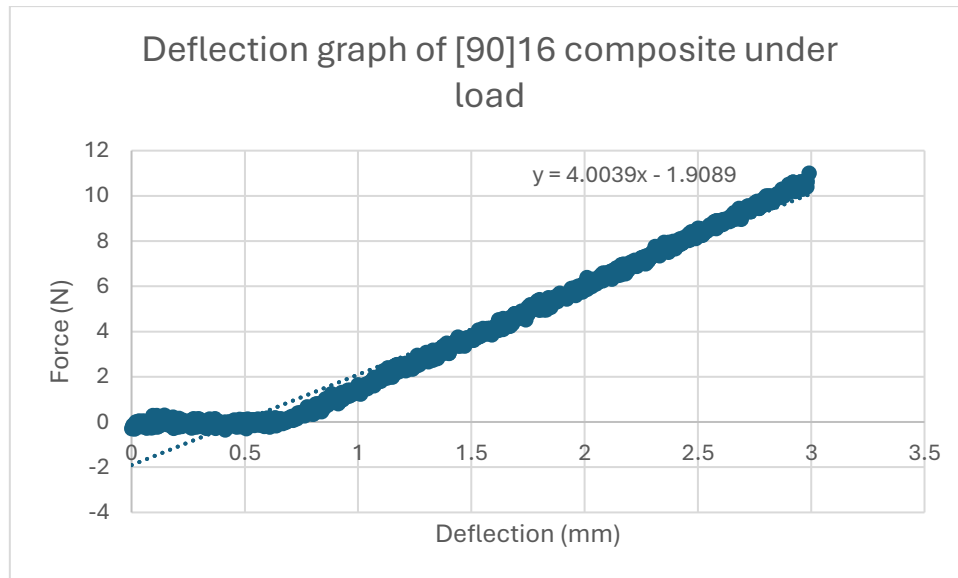
$$d = 2.18 \times 10^{-3}m$$

$$E = 75,358 \cdot \frac{0.122^3}{4 \cdot 0.02592 \cdot 2.18 \times 10^{-3^3}}$$

$$E = 1.27 \times 10^{11} Pa \quad \mathbf{127GPa}$$

6.2 Experimental Calculations for Test 2

Figure 5: Deflection graph for $[90]_{16}$



$$E = slope \cdot \frac{L^3}{4bd^3}$$

$$Slope = 4,003.9N/m$$

$$L = 0.122m$$

$$b = 0.02577m$$

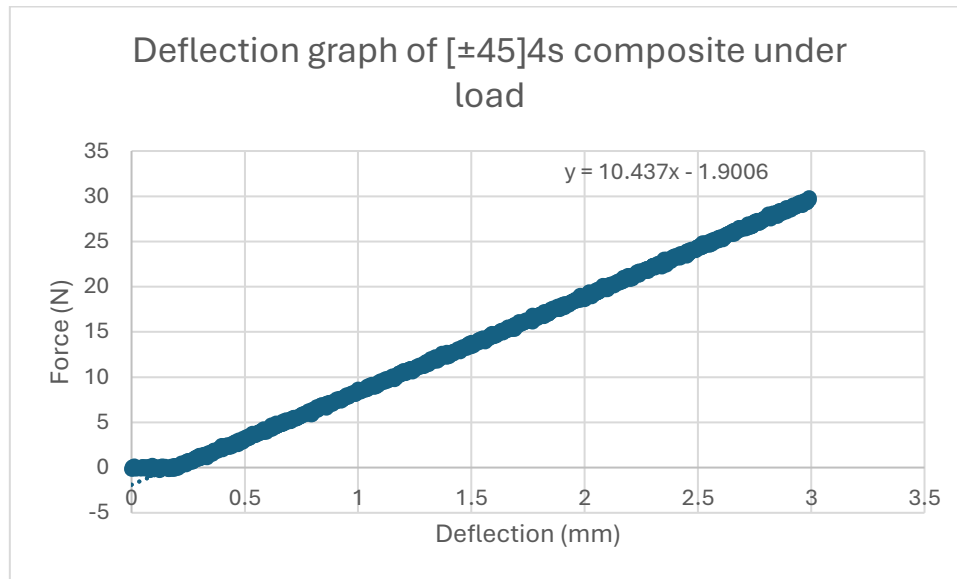
$$d = 2.17 \times 10^{-3}m$$

$$E = 4,003.9 \cdot \frac{0.122^3}{4 \cdot 0.02577 \cdot 2.17 \times 10^{-3^3}}$$

$$E = 6.90 \times 10^9 Pa \quad \mathbf{6.90GPa}$$

6.3 Experimental Calculations for Test 3

Figure 6: Deflection graph for $[\pm 45^\circ]_{4s}$



$$E = \text{slope} \cdot \frac{L^3}{4bd^3}$$

$$\text{Slope} = 10,437 \text{ N/m}$$

$$L = 0.122 \text{ m}$$

$$b = 0.02504 \text{ m}$$

$$d = 2.22 \times 10^{-3} \text{ m}$$

$$E = 10,437 \cdot \frac{0.122^3}{4 \cdot 0.02504 \cdot 2.22 \times 10^{-3^3}}$$

$$E = 1.729 \times 10^{10} \text{ Pa} \quad \mathbf{17.29 \text{ GPa}}$$

Table 7: Summary of experimental results

Sample No.	Stacking sequence	Theoretical Modulus (GPa)	Experimental Modulus (GPa)
1	$[0]_{16}$	125	127
2	$[90]_{16}$	12.5	6.9
3	$[\pm 45/-45]_{4s}$	12.5	17.3

7. Results Comparison

7.1 Comparison between $[0]_{16}$ (M40J/Epoxy) and $[0]_{16}$ (CFRP)

The modulus calculated for the proposed $[0]_{16}$ composite is 227.52GPa and the $[0]_{16}$ laboratory sample has a theoretical modulus of 125GPa. The modulus differ as M40J is a premium carbon fibre, it is expensive which will make it unsuitable for use in a university laboratory session.

Structural Perspective –

- Proposed $[0]_{16}$ ($E_c = 227.5\text{GPa}$): The proposed M40J/Epoxy resin composite achieves a much higher modulus than the laboratory sample. This means for the application of the F1 front wing the proposed design would be much more favourable as it would be much stiffer and experience much less deflection. This stiffness does mean that this composite will be more brittle. This is important for aerodynamic stability and complying with strict FIA regulations.
- Laboratory Sample $[0]_{16}$ ($E_c = 125\text{GPa}$): This sample is less stiff and would experience more deformation under the same load as the proposed design, this means it would offer worse aerodynamic performance and compliance with the regulations.

Cost Perspective –

- Proposed $[0]_{16}$: M40J is a very premium carbon fibre coming at a high cost (\$130 - \$150 per kg) [16] due to its impressive mechanical properties. Epoxy resin (\$20-\$40 for 500ml) [22] also has cheaper alternatives like polyester, but due to its strong adhesive properties to carbon it is needed for this design. High modulus fibres like M40J also come with more experience manufacturing techniques in the form of equipment or specialised tools like autoclaves. Also, high modulus carbon fibres aren't sustainable as not much recyclable technology can be used, however as discussed research is ongoing to reduce this in the future.
- Laboratory Sample $[0]_{16}$: The lab sample will have contained less premium carbon fibre. Therefore, reducing the overall modulus of the composite. However, it will be more accessible and more cost sensitive. Even in manufacturing the process would be cheap due to less sensitive process environments. This composite is more sustainable as it isn't as stiff, however overall sustainability is limited with all carbon fibres.

Advantages and Disadvantages –

Table 8: Summary of advantages and disadvantages of proposed and laboratory composites

Composite	Advantages	Disadvantages
Proposed Composite	High stiffness	A lot more expensive, more brittle, more complex manufacturing process and poor sustainability
Laboratory sample	Suitable stiffness, more cost effective and easier to manufacture	Lower stiffness, more deflection and poor sustainability

7.2 Results of three samples tested during laboratory session

Table 9: Summary of overall results

Sample No.	Stacking sequence	Theoretical Modulus (GPa)	Experimental Modulus (GPa)	Difference (GPa)
1	$[0]_{16}$	125	127	+2 (1.6%)
2	$[90]_{16}$	12.5	6.9	-5.6 (44.8%)
3	$[+45/-45]_{4s}$	12.5	17.3	+4.8 (38.4%)

- A positive % difference means the experimental modulus was higher than predicted.
- A negative % difference means the experimental modulus was lower than predicted.

$[0]_{16}$ Laminate –

The theoretical and experimental modulus are very close, this indicated that the laminate has good fibre alignment and matrix quality. The high modulus is due to all the fibre being orientated with the direction of the load therefore, the carbon fibres are bearing the most load.

$[90]_{16}$ Laminate –

The experimental modulus is significantly lower than the theoretical value. This laminate has fibres that are perpendicular to the load, so the matrix is bearing most of the load, meaning the laminate has a low modulus. The difference in the values could be due to imperfect bonding, matrix defects, and wear on the laminate due to repeated bend cycles.

$[+45/-45]_{4s}$ Laminate –

The experimental modulus is higher than the theoretical value. This could be due to the 10% rule not fully capturing the fact that fibres at 45° will offer some load transfer into the composites. A way to improve this in future work would be to use advanced laminate theory.

In summary, the 10% Rule simplifies and estimates that any off axis piles will only offer 10% of the axial modulus. However, as the 45° fibres are close the direction of the load a small contribution from the fibre will be made. This highlights the need for more advanced analysis. More advanced theories include Classical laminate theory and the Tsai hill failure criterion, both provide more accurate theoretical approach as they account for ply orientation, and interlaminar stresses. On top of this, micromechanical models take defects into consideration e.g. voids which will allow for a more accurate calculation and results than the rule of mixture and 10% rule. [32] [33]

Sources of Error –

- Defects: such as voids or fibre misalignment can result in a lower modulus
- Imperfect bonding: a poor fibre matrix bond will cause an inefficient load transfer, this will cause the modulus to decrease
- Measurement uncertainty: there could be errors with the measurement recording readings during the test.

8. Conclusion

This report investigated the design and analysis of composite laminates for formula one applications, which were focused on the front wing. The proposed M40J/Epoxy composite achieved a theoretical Voigt Modulus of 227.5 GPa, this outperformed a similar composite that we tested in the laboratory session. This composite had theoretical modulus of 125 GPa and an experimental modulus of 127 GPa.

Comparing the theoretical and experimental results for all sequences of the same composite properties, shows a very close relation between theoretical and experimental for the $[0]_{16}$ however, for the off axis laminates ($[90]_{16}$ and $[+45/-45]_{4s}$), there were larger differences of around $\pm 40\%$. These results are due to the simplifying of the 10% rule, which does consider partially of axis fibres as it assumes all fibres that are parallel to the direction of the force only add 10% of the stiffness of the fibres in parallel. Also, there was a difference between all theoretical and experimental results as theoretical calculations don't capture manufacturing defects, imperfect bonding and measurement uncertainties.

This work shows the importance of using both theoretical and experimental analysis in materials engineering. Theoretical calculations can display valuable rough predictions, experimental testing is required as it accounts for real world variables and effects.

For future work, it could be useful to explore more advanced composite theory and do more research into hybrid composites which could be the future of composites within formula one and material science.

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